

**BAULKHAM HILLS
HIGH
SCHOOL**

2020

**YEAR 12
ASSESSMENT TASK 3**

Mathematics Extension 2

General

- Reading time – 5 minutes

Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black or blue pen
Black pen is preferred
- Board- approved calculators may be used
- In Questions 6-8, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks:
47

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section

Section II – 42 marks (pages 4 – 8)

- Attempt Questions 6-8
- Allow about 82 minutes for this section

Section I

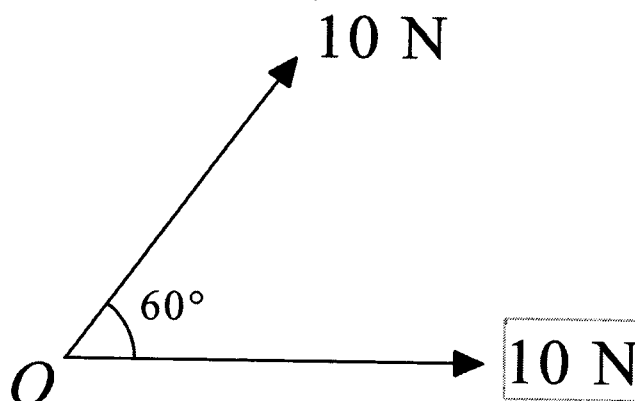
5 marks

Attempt questions 1-5

Allow about 8 minutes for this section

Use the multiple choice answer sheet in your booklet for Questions 1 – 5.

1. A particle is moving in simple harmonic motion which satisfies $\ddot{x} = -4x$.
The particle starts from $x = 0$ with initial velocity 4m/s.
Which of the following is a possible equation for the displacement, x , of the particle?
- (A) $x = 2 \sin 2t$
- (B) $x = 2 \sin 4t$
- (C) $x = 4 \sin 2t$
- (D) $x = 4 \sin 4t$
2. Two forces of magnitude 10 N act on a particle at O as shown. What is the magnitude of the resultant force?



- (A) 5
- (B) 10
- (C) $10\sqrt{3}$
- (D) 20

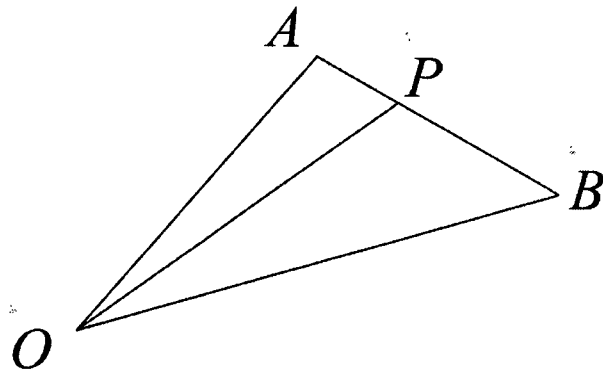
3. The velocity, v m/s of a particle moving in a straight line is governed by the equation $v = x - 3$, where x is the displacement of the particle at time t seconds. Initially the particle was at $x = 7$. What is the displacement function of this particle?

- (A) $x = 7 + e^t$
 (B) $x = 7e^t$
 (C) $x = 3 + e^t$
 (D) $x = 3 + 4e^t$

4. In this diagram, $\vec{OA} = 6\vec{i} - \vec{j} + 8\vec{k}$, $\vec{OB} = -3\vec{i} + 4\vec{j} - 2\vec{k}$ and $AP:PB = 1:2$.

The vector $\vec{OP}$ is equal to

- (A) $\frac{7}{3}\vec{i} + \frac{4}{3}\vec{k}$
 (B) $3\vec{i} + \frac{7}{3}\vec{j} + \frac{4}{3}\vec{k}$
 (C) $3\vec{j} + 4\vec{k}$
 (D) $3\vec{i} + \frac{2}{3}\vec{j} + \frac{14}{3}\vec{k}$



5. A particle of mass m is moving in a straight line under the action of a force where

$$F = \frac{m}{x^3}(6 - 10x).$$

Which of the following is an equation for its velocity at any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{-3 + 10x - 7x^2}$
 (B) $v = \pm x \sqrt{-3 + 10x - 7x^2}$
 (C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$
 (D) $v = \pm x \sqrt{2(-3 + 10x - 7x^2)}$

Section II

42 marks

Attempt questions 6-8

Allow about 82 minutes for this section

Answer each question on the appropriate page of your answer booklet

In Questions 6-8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks) Answer on the appropriate page

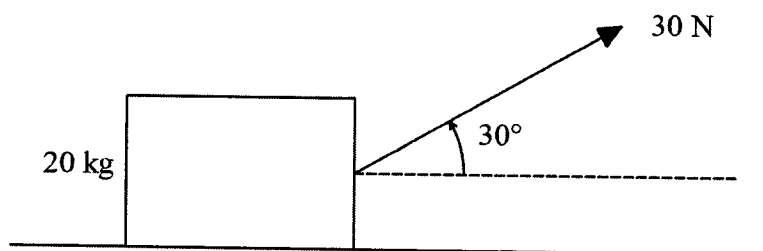
(a) Given $P = (2, 3, -1)$ and $Q = (-4, 4, 2)$, find

- | | | |
|------|----------------------------------|---|
| i) | $\overrightarrow{PQ}$ | 1 |
| ii) | The distance between P and Q | 1 |
| iii) | The midpoint of PQ | 1 |

(b) If $\underline{u} = \begin{bmatrix} -4 \\ 2 \\ 1 \end{bmatrix}$ and $\underline{v} = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$, find

- | | | |
|-----|---|---|
| i) | $\underline{u} \cdot \underline{v}$ | 1 |
| ii) | The angle between $\underline{u}$ and $\underline{v}$ (correct to the nearest minute) | 2 |

(c) A box of mass 20 kg is pulled along a smooth horizontal table by a force of 30 newtons acting at an angle of 30° to the horizontal.



Find the magnitude of the normal reaction of the table on the box. Use $g = 10 \text{ m/s}^2$.

2

Question 6 continues on the following page

(d) Find the values of k given that $\begin{bmatrix} 2k \\ \frac{1}{\sqrt{2}} \\ -k \end{bmatrix}$ is a unit vector. **2**

(e) Find all vectors which are perpendicular to both $\begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ **3**

End of Question 6

Question 7 (13 marks) Answer on the appropriate page

- (a) Find the projection of $\underline{a}$ onto $\underline{b}$ given $\underline{a} = 3\underline{i} - \underline{j} + 2\underline{k}$ and $\underline{b} = \underline{i} + 3\underline{j} + 5\underline{k}$ **2**
- (b) (i) Find the vector equation of the line through the points (3,-1,-2) and (5,3,-4) **2**
- (ii) Find the point(s) where this line meets the sphere $x^2 + y^2 + z^2 = 26$. **2**
- (c) A particle moves so that its displacement, x cm, from the origin, O , at time t seconds is given by $x = \sqrt{3} \cos 3t - \sin 3t$.
- i) Show that the particle moves in simple harmonic motion. **2**
- ii) Find the period of the motion. **1**
- iii) Find the time when the particle first passes through the origin. **2**
- iv) What is the particle's velocity the first time it is 1 cm from the origin? **2**

End of Question 7

Question 8 (16 marks) Answer on the appropriate page

- (a) A particle of mass m kilograms is falling from rest. It experiences air resistance of magnitude mkv^2 where k is a positive constant and v is the velocity of the particle.

The terminal velocity of the particle during the downward motion is V_T .

- (i) By considering the forces on the particle during its downward motion, show that $kV_T^2 = g$ (where g is the acceleration due to gravity). 2

- (ii) The particle was then projected vertically upwards from level ground. The initial speed of projection is $\frac{1}{5}$ of the magnitude of the terminal velocity from the downward motion. 2

Show that during its upward motion the acceleration of the particle, $\ddot{x}$, is

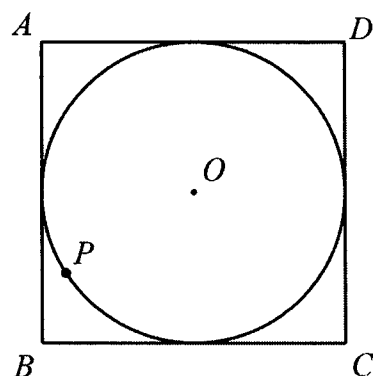
given by
$$\ddot{x} = -g \left(1 + \frac{v^2}{(V_T)^2} \right)$$

- (iii) Show that the maximum height, H , reached is given by 3

$$H = \frac{V_T^2}{2g} \ln \left(\frac{26}{25} \right).$$

Question 8 continues on the following page

- (b) In the figure, the circle has centre O and radius r . The circle is inscribed in a square $ABCD$ and P is any point on the circle.



- i) Show that $\vec{AP} \cdot \vec{AP} = 3r^2 - 2\vec{OP} \cdot \vec{OA}$ 2
- ii) Hence find $AP^2 + BP^2 + CP^2 + DP^2$ in terms of r . 2

- (c) A short-barrelled machine gun stands on horizontal ground. The gun fires bullets, from ground level, at speed u continuously from $t = 0$ to $t = \frac{\pi}{6\lambda}$, where λ is a positive constant. During this time period, the angle of elevation α of the barrel decreases from $\frac{\pi}{3}$ to $\frac{\pi}{6}$ and is given at time t by $\alpha = \frac{\pi}{3} - \lambda t$.

You may assume the equations of motion $x = ut \cos \alpha$ and $y = -\frac{gt^2}{2} + ut \sin \alpha$

- (i) T is the time it takes for a bullet, fired t seconds after the first bullet, 2
to hit the ground. Show that $T = t + \frac{2u}{g} \sin \left(\frac{\pi}{3} - \lambda t \right)$.
- (ii) Show that the last bullet to hit the ground does so at a distance 3

$$\frac{2ku^2 \sqrt{1-k^2}}{g} \text{ from the gun, where } k = \frac{g}{2\lambda u}.$$

End of paper

EXTENSION 2 TASK 3 2020

ML. 1. $n^2 = 4$

$n = 2$

Consider (A): $x = 2 \sin 2t$ (For B: $x = 8 \cos 4t$)
 $\dot{x} = 4 \cos 2t$ $\dot{x}(0) = 8$

when $t=0$, $x=0$

$\dot{x} = 4$

$\therefore (A)$

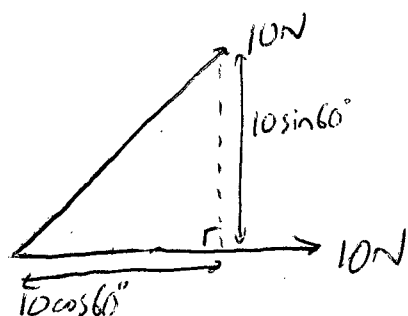
C: $\dot{x} = 8 \cos 2t$

$\dot{x} = 8$

D: $\dot{x} = 16 \cos 4t$

$\dot{x} = 16$)

2.



$\therefore$ Resultant $10 \hat{i} + 10 \cos 60^\circ \hat{j} + 10 \sin 60^\circ \hat{j}$

$= 15 \hat{i} + 5\sqrt{3} \hat{j}$

Magnitude $= \sqrt{15^2 + (5\sqrt{3})^2}$

$= \sqrt{300}$

$= 10\sqrt{3} \text{ N}$

$\therefore (C)$

3.

$\frac{dn}{dt} = n-3$

$\frac{dt}{dn} = \frac{1}{n-3}$

$\int_0^t dt = \int_4^n \frac{1}{n-3} dn$

$[t]_0^t = [\ln|n-3|]_4^n$

$t-0 = \ln|n-3| - \ln 4$

$t = \ln \left| \frac{n-3}{4} \right|$

$\frac{n-3}{4} = e^t$

$n-3 = 4e^t$

$n = 3 + 4e^t$

$\therefore (D)$

4.

$$\begin{aligned}\vec{OP} &= \vec{OA} + \frac{1}{112} \vec{AB} \\ &= 6\hat{i} - 2\hat{j} + 8\hat{k} + \frac{1}{3}((-3-6)\hat{i} + (4-1)\hat{j} + (-2-8)\hat{k}) \\ &= 6\hat{i} - 2\hat{j} + 8\hat{k} + \frac{1}{3}(-9\hat{i} + 3\hat{j} - 10\hat{k})\end{aligned}$$

$$\vec{OP} = 3\hat{i} + \frac{2}{3}\hat{j} + \frac{14}{3}\hat{k} \quad \therefore (D)$$

5.

$$F = \frac{m}{n^3} (6 - 10n)$$

$$ma = \frac{m}{n} (6 - 10n)$$

$$\frac{d}{dn} \left(\frac{1}{2} V^2 \right) = 6n^{-3} - 10n^{-2}$$

$$\frac{V^2}{2} = -3n^{-2} + 10n^{-1} + C$$

When $n=1, V=0$

$$0 = 7 + C$$

$$C = -7$$

$$\therefore V^2 = \frac{-6}{n^2} + \frac{20}{n} - 14$$

$$V^2 = \frac{1}{n^2} (2(-3 + 10n - 7n^2))$$

$$V = \pm \frac{1}{n} \sqrt{2(-3 + 10n - 7n^2)}$$

$$\therefore (C)$$

$$b) a) i) \vec{PQ} = \begin{bmatrix} -4 \\ 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \\ = \begin{bmatrix} -6 \\ 1 \\ 3 \end{bmatrix}$$

$$ii) |\vec{PQ}| = \sqrt{(-6)^2 + 1^2 + 3^2} \\ = \sqrt{46}$$

$$iii) \text{Midpoint} = \left(-\frac{3}{2}, \frac{7}{2}, \frac{1}{2}\right) \\ = \left(-1, \frac{7}{2}, \frac{1}{2}\right)$$

① correct answer
(can't be expressed as coordinates)

① correct answer

① correct answer

$$b) i) \underline{u} \cdot \underline{v} = (-4 \times -1) + (2 \times 3) + (1 \times 2) \\ = 12$$

$$ii) \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \\ = \frac{12}{\sqrt{21} \times \sqrt{14}}$$

$$= \frac{12}{7\sqrt{6}} \text{ or } \frac{12\sqrt{6}}{7}$$

$$\therefore \theta = \cos^{-1} \frac{12}{7\sqrt{6}}$$

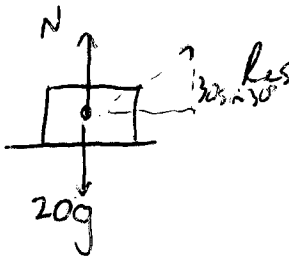
$$\theta = 45^\circ 35' \text{ nearest minute}$$

① correct answer

② correct answer

① correct expression for $\cos \theta$

c) i)



Resting vertically:

$$N + 30 \sin 30^\circ = 20g$$

$$N + 30 \times \frac{1}{2} = 200$$

$$N = 185 \text{ N.}$$

② correct solution

① Resolves force into vertical component

6 d)

$$\sqrt{(2k)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + (-k)^2} = 1$$

$$5k^2 + \frac{1}{2} = 1$$

$$5k^2 = \frac{1}{2}$$

$$k^2 = \frac{1}{10}$$

$$k = \pm \frac{1}{\sqrt{10}}$$

② Correct solution

① Attempts to find magnitude and equate to 1

e) let perpendicular vector be $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$. Since perpendicular, dot products must be 0.

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix} = 0$$

$$2a - b - 2c = 0 \quad ①$$

$$\therefore ① \times 2$$

$$3a = 0$$

$$a = 0$$

$$\text{From ① } b = -2c$$

$$\text{and } \begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 0$$

$$a + b + 2c = 0 \quad ②$$

③ correct solution

② finds $a=0$ and relationship between b and c

① finds $a=0$ or equates dot products to be 0.

$$\text{from ② } b = -2c$$

$\therefore$ vector is of the form: $\lambda \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$ will be perpendicular.

7 a) $\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{b} \cdot \underline{a}}{|\underline{b}|^2} \underline{b}$

$$= \frac{3 \times 1 - 1 \times 3 + 5 \times 2}{1 + 3 + 5^2} (\underline{i} + 3\underline{j} + 5\underline{k})$$

$$= \frac{10}{35} (\underline{i} + 3\underline{j} + 5\underline{k})$$

$$= \frac{2}{7} \underline{i} + \frac{6}{7} \underline{j} + \frac{10}{7} \underline{k}$$

② Correct solution

① progress towards (correct formula with one error in calculation)

7 b) i) Equation of line through points is

$$\underline{r} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 5-3 \\ 3-(-1) \\ -4+2 \end{pmatrix} \quad \text{where } \lambda \text{ is a constant}$$

$$\underline{r} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix}$$

$$\text{or } \underline{r} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$\lambda_1 = 2\lambda$
not penalised if omitted

- (3) correct solution
- (2) forms correct quadratic equation to find λ
- (1) attempts to find point of intersection by substituting coordinates of point on the line into equation of sphere

ii)

$$\underline{r} = \begin{pmatrix} 3+\lambda \\ -1+2\lambda \\ -2-\lambda \end{pmatrix}$$

Since $\underline{r}$ lies on sphere when they intersect.

$$\begin{aligned} (3+\lambda)^2 + (-1+2\lambda)^2 + (-2-\lambda)^2 &= 26 \\ \lambda^2 + 6\lambda + 9 + 4\lambda^2 - 4\lambda + 1 + \lambda^2 + 4\lambda + 4 &= 26 \\ 6\lambda^2 + 6\lambda + 14 &= 26 \end{aligned}$$

$$6\lambda^2 + 6\lambda - 12 = 0$$

$$\lambda^2 + \lambda - 2 = 0$$

$$(\lambda+2)(\lambda-1) = 0$$

$$\lambda = -2: \text{ Pt is } (1, -5, 0)$$

$$\lambda = 1: \text{ Pt is } (4, 1, -3)$$

$\therefore$ Points of intersection are $(1, -5, 0)$ and $(4, 1, -3)$

7 c) i)

$$\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$= -9(\sqrt{3} \cos 3t - \sin 3t)$$

$$\ddot{x} = -9x \text{ which is of the form } \ddot{x} = -n^2x$$

$$\text{where } n=3$$

ii)

$$T = \frac{2\pi}{n}$$

$$n^2 = 9$$

$$\text{Period} = \frac{2\pi}{3} \text{ seconds}$$

$$n=3$$

- (2) correct solution
- (1) finds acceleration

(1) correct answer

7 c iii) When $x=0$, $0 = \sqrt{3} \cos 3t - \sin 3t$
 $\sin 3t = \sqrt{3} \cos 3t$
 $\tan 3t = \sqrt{3}$
 $3t = \frac{\pi}{3}$
 $t = \frac{\pi}{9}$ seconds

② correct answer (must be exact)

① obtains trig equation
 $\tan(3t) = \sqrt{3}$ or
 equivalent

c iv) let $\sqrt{3} \cos 3t - \sin 3t = \sqrt{3+1} \cos(3t + \alpha)$
 $\cos \alpha = \frac{\sqrt{3}}{2}$
 $\alpha = \frac{\pi}{6}$

② correct solution

① significant progress
 (eg expresses displacement
 as $2(\cos 3t + \frac{\pi}{6})$)

When $x=1$,

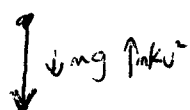
$\therefore 2 \cos(3t + \frac{\pi}{6}) = 1$
 $\cos(3t + \frac{\pi}{6}) = \frac{1}{2}$
 $3t + \frac{\pi}{6} = \frac{\pi}{3}$
 $3t = \frac{\pi}{6}$
 $t = \frac{\pi}{18}$

$\therefore$ passes through 1 when $t = \frac{\pi}{18}$

$v = -3\sqrt{3} \sin \frac{\pi}{6} - 3 \cos \frac{\pi}{6}$
 $= -\frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2}$

$\therefore$ Velocity = $-3\sqrt{3}$ cm/s

8 a) i) Downward



$F = ma$

for terminal velocity $a=0$

$mg - mkv_t^2 = 0$

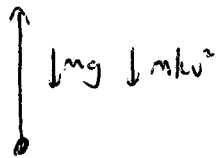
$kv_t^2 = g$ as req'd

② correct solution

① correct force diagram
 (or correct equation for net
 force)

8 a ii)

upward



$$ma = -mg - mkv^2$$

$$a = -(g + kv^2)$$

But from i) $g = kV_T^2$

$$\ddot{x} = -\left(\frac{gV^2}{V_T^2} + g\right)$$

$$\ddot{x} = -g\left(1 + \frac{V^2}{V_T^2}\right)$$

iii)

Using $\ddot{x} = v \frac{dv}{dx}$

$$v \frac{dv}{dx} = -g \left(\frac{V^2 + V_T^2}{V_T^2} \right)$$

$$\int_0^H dx = -\frac{V_T^2}{g} \int_0^{V_T/5} \frac{v dv}{V^2 + V_T^2}$$

$$[x]_0^H = -\frac{V_T^2}{2g} \left[\ln(V^2 + V_T^2) \right]_0^{V_T/5}$$

$$H - 0 = \frac{V_T^2}{2g} \left[\ln(V^2 + V_T^2) \right]_0^{V_T/5}$$

$$= \frac{V_T^2}{2g} \ln \left(\frac{\frac{V_T^2}{25} + V_T^2}{V_T^2} \right)$$

$$= \frac{V_T^2}{2g} \ln \left(\frac{26 V_T^2}{25 V_T^2} \right)$$

$$= \frac{V_T^2}{2g} \ln \left(\frac{26}{25} \right)$$

② correct proof

① finds correct expression for $\ddot{x}$ in terms of g, k, v

③ correct solution

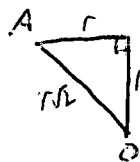
② significant progress (eg correctly integrates excluding limits)

① uses $v \frac{dv}{dx}$ and expresses as correct integral.

8 b) i)

$$\vec{OA} + \vec{AP} = \vec{OP}$$

$$\vec{AP} = \vec{OP} - \vec{OA}$$



$$|\vec{OA}| = |\vec{OB}| = |\vec{OC}| = |\vec{OD}| = r/2$$

Using cosine rule in $\triangle AOP$

$$|\vec{AP}|^2 = |\vec{OA}|^2 + |\vec{OP}|^2 - 2|\vec{OA}||\vec{OP}|\cos(\angle AOP)$$

$$|\vec{AP}|^2 = 2r^2 + r^2 - 2|\vec{OA}||\vec{OP}| \frac{\vec{OA} \cdot \vec{OP}}{|\vec{OA}||\vec{OP}|}$$

$$|\vec{AP}|^2 = 3r^2 - 2\vec{OA} \cdot \vec{OP}$$

$$\vec{AP} \cdot \vec{AP} = 3r^2 - 2\vec{OA} \cdot \vec{OP}$$

$$\text{since } \vec{r} \cdot \vec{r} = |\vec{r}|^2$$

ii) Similarly

$$BP^2 = \vec{BP} \cdot \vec{BP} = 3r^2 - 2\vec{OB} \cdot \vec{OP}$$

$$CP^2 = \vec{CP} \cdot \vec{CP} = 3r^2 - 2\vec{OC} \cdot \vec{OP}$$

$$DP^2 = \vec{DP} \cdot \vec{DP} = 3r^2 - 2\vec{OD} \cdot \vec{OP}$$

$$AP^2 + BP^2 + CP^2 + DP^2 = 12r^2 - 2(\vec{OA} \cdot \vec{OP} + \vec{OB} \cdot \vec{OP} + \vec{OC} \cdot \vec{OP} + \vec{OD} \cdot \vec{OP})$$

But diagonals of a square bisect $\vec{OB} = -\vec{OA}$ and $\vec{OC} = -\vec{OD}$

$$\therefore AP^2 + BP^2 + CP^2 + DP^2 = 12r^2 - 2(\vec{OA} \cdot \vec{OP} - \vec{OA} \cdot \vec{OP} + \vec{OP} \cdot -\vec{OB} + \vec{OP} \cdot \vec{OB})$$

$$\therefore AP^2 + BP^2 + CP^2 + DP^2 = 12r^2$$

② correct solution*

① uses cosine rule in $\triangle OPA$ or dot product
(note * must define θ)

② correct solution

① finds expression for sum of squares

8 < i)

Time of flight

$$y = -gt^2 + ut \sin \alpha = 0$$

$$gt^2 - 2ut \sin \alpha = 0$$

$$t(gt - 2u \sin \alpha) = 0$$

$$t = 0 \text{ or } t = \frac{2u \sin \alpha}{g}$$

$$\therefore \text{time of flight } \frac{2u \sin \alpha}{g}$$

$$\text{Time since: } T_L = t + \frac{2u \sin \alpha}{g}$$

$$T_L = t + \frac{2u}{g} \sin\left(\frac{\pi}{3} - \lambda t\right)$$

ii) For last bullet to hit the ground, T_L will be a maximum

$$\frac{dT_L}{dt} = 1 - \frac{2u\lambda}{g} \cos\left(\frac{\pi}{3} - \lambda t\right)$$

$$= 1 - \frac{1}{k} \cos\left(\frac{\pi}{3} - \lambda t\right)$$

$$\text{For max value } \frac{dT_L}{dt} = 0$$

$$1 - \frac{1}{k} \cos\left(\frac{\pi}{3} - \lambda t\right) = 0$$

$$k - \cos\left(\frac{\pi}{3} - \lambda t\right) = 0$$

$$k = \cos\left(\frac{\pi}{3} - \lambda t\right) *$$

$$\frac{d^2 T_L}{dt^2} = -\frac{2\lambda^2 u}{g} \sin\left(\frac{\pi}{3} - \lambda t\right)$$

$$< 0 \therefore \text{max value}$$

$$\text{Horizontal range, } R = ut \cos \alpha$$

$$= u \frac{2u \sin \alpha}{g} \cos \alpha$$

$$\text{Range} = \frac{2u^2 \sin \alpha \cos \alpha}{g}$$

$$\text{From } * \quad k = \cos \alpha, \quad \sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - k^2}$$

$$\therefore \text{Range} = \frac{2ku^2 \sqrt{1 - k^2}}{g} \text{ as req'd.}$$

(2) correct solution

(1) calculates time of flight

(3) correct solution

(2) finds range in terms of u, α, g and obtains $k = \cos\left(\frac{\pi}{3} - \lambda t\right)$

(1) Finds expression for range in terms of u, α, g or (1) attempts to find maximum time by differentiating and solving